

Center of Mass

A beginner-friendly physics guide

Key idea: the center of mass is the mass-weighted average position of an object or a group of objects. Heavier parts pull this average closer to them.

1. What does it mean?

Imagine balancing a ruler on one finger. A uniform ruler balances at its midpoint. Attach a small weight to one end and the balance point moves toward that end. In approximately uniform gravity, this balance point corresponds to the center of mass.

The center of mass is a position, not a special piece of material. For a uniform ring, it lies at the empty center.

2. The formula

$$x_{\text{CM}} = (m_1 x_1 + m_2 x_2 + \dots) / (m_1 + m_2 + \dots)$$

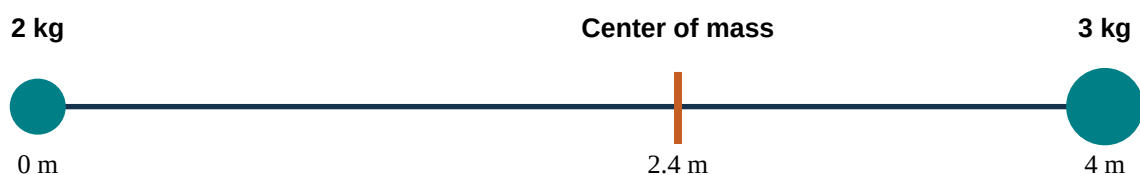
m is mass; **x** is position measured from one chosen origin. Use the same position units throughout. The answer has units of length. For two or three dimensions, apply the same formula separately to x, y, and z.

Method: choose an origin; multiply each mass by its position; add these products; divide by total mass.

3. Worked example: two masses

A 2 kg mass is at $x = 0$ m and a 3 kg mass is at $x = 4$ m. Treat both as point masses and ignore any connecting rod.

$$x_{\text{CM}} = (2 \times 0 + 3 \times 4) / (2 + 3) = 2.4 \text{ m}$$



The answer is closer to the heavier, 3 kg mass: 1.6 m from it and 2.4 m from the 2 kg mass. Check: $2 \times 2.4 = 3 \times 1.6$.

4. Common shapes

For uniform mass distributions: a straight rod has its center of mass at its midpoint; a rectangular plate at the intersection of its diagonals; a disk, ring, or sphere at its geometric center.

Examples & Quick Revision

5. A rod with an attached mass

A uniform 2 m rod has mass 4 kg. A 2 kg point mass is attached at its right end. Measure x from the left end. The rod's own center of mass is at $x = 1$ m; the attached mass is at $x = 2$ m.

$$x_{\text{CM}} = (4 \times 1 + 2 \times 2) / (4 + 2) = 1.33 \text{ m}$$

When combining objects, treat each object's mass as located at its own center of mass. Do not forget the rod's mass.

6. Center of mass in two dimensions

Place 1 kg at (0, 0), 1 kg at (4, 0), and 2 kg at (0, 3), with coordinates in meters.

$$x_{\text{CM}} = (1 \times 0 + 1 \times 4 + 2 \times 0) / 4 = 1 \text{ m}$$
$$y_{\text{CM}} = (1 \times 0 + 1 \times 0 + 2 \times 3) / 4 = 1.5 \text{ m}$$

The center of mass is **(1, 1.5) m**. Use the same total mass for both coordinate calculations.

7. Why it matters

Motion: for a constant-mass system, total external force equals total mass times center-of-mass acceleration: $F_{\text{ext}} = M a_{\text{CM}}$. Zero net external force means constant center-of-mass velocity, not necessarily zero velocity.

Balance: under gravity, a stationary object on a level surface tends to tip when the vertical line through its center of mass passes outside its support area. A lower center of mass and wider base generally make tipping harder.

Center of gravity: coincides with center of mass in a uniform gravitational field; the two need not coincide when gravity varies significantly.

8. Practice

Question	Answer
Equal masses at $x = 1$ m and $x = 5$ m.	3 m
1 kg at $x = 0$ m and 3 kg at $x = 8$ m.	6 m
1 kg at 0 m, 2 kg at 2 m, and 3 kg at 4 m.	8/3 m (about 2.67 m)

Avoid these mistakes: taking a simple position average when masses differ; mixing origins; ignoring an object's own mass; assuming the center of mass must lie inside solid material.